Substitution Theorem

The next theorem provides the justification for the "change of variable" method that is often used to evaluate integrals. This theorem is employed (usually implicitly) in the evaluation by means of procedures that involve the manipulation of "differentials," common in elementary courses.

7.3.8 Substitution Theorem *Let $J := [\alpha, \beta]$ and let $\varphi : J \to \mathbb{R}$ have a continuous derivative on J. If $f : I \to \mathbb{R}$ is continuous on an interval I containing $\varphi(J)$, then*

$$\int_{\alpha}^{\beta} f(\varphi(t)) \cdot \varphi'(t)dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)dx. \tag{5}$$

The proof of this theorem is based on the Chain Rule 6.1.6, and will be outlined in Exercise 17. The hypotheses that f and φ' are continuous are restrictive, but are used to ensure the existence of the Riemann integral on the left side of (5).

7.3.9 Examples **(a)** Consider the integral $\displaystyle\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}dt$.

Here we substitute $\varphi(t) := \sqrt{t}$ for $t \in [1, 4]$ so that $\varphi'(t) = 1/(2\sqrt{t})$ is continuous on [1, 4]. If we let $f(x) := 2\sin x$, then the integrand has the form $(f \circ \varphi) \cdot \varphi'$ and the Substitution Theorem 7.3.8 implies that the integral equals $\int_1^2 2\sin x\,dx = -2\cos x|_1^2 = 2(\cos 1 - \cos 2)$.

(b) Consider the integral $\displaystyle\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}dt$.

Since $\varphi(t) := \sqrt{t}$ does not have a continuous derivative on [0, 4], the Substitution Theorem 7.3.8 is not applicable, at least with this substitution. (In fact, it is not obvious that this integral exists; however, we can apply Exercise 7.2.11 to obtain this conclusion. We could then apply the Fundamental Theorem 7.3.1 to $F(t) := -2\cos\sqrt{t}$ with $E := \{0\}$ to evaluate this integral.) □

We will give a more powerful Substitution Theorem for the *generalized* Riemann integral in Section 10.1.

Lebesgue's Integrability Criterion

We will now present a statement of the definitive theorem due to Henri Lebesgue (1875–1941) giving a necessary and sufficient condition for a function to be Riemann integrable, and will give some applications of this theorem. In order to state this result, we need to introduce the important notion of a null set.

Warning Some people use the term "null set" as a synonym for the terms "empty set" or "void set" referring to $\emptyset$ (= the set that has no elements). However, we will always use the term "null set" in conformity with our next definition, as is customary in the theory of integration.

7.3.10 Definition **(a)** A set $Z \subset \mathbb{R}$ is said to be a **null set** if for every $\varepsilon > 0$ there exists a countable collection $\{(a_k, b_k)\}_{k=1}^{\infty}$ of open intervals such that

$$Z \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k) \qquad \text{and} \qquad \sum_{k=1}^{\infty}(b_k - a_k) \leq \varepsilon. \tag{6}$$

(b) If $Q(x)$ is a statement about the point $x \in I$, we say that $Q(x)$ holds **almost everywhere** on I (or for **almost every** $x \in I$), if there exists a null set $Z \subset I$ such that $Q(x)$ holds for all $x \in I \backslash Z$. In this case we may write

$$Q(x) \quad \text{for a.e.} \quad x \in I.$$

It is trivial that any subset of a null set is also a null set, and it is easy to see that the union of two null sets is a null set. We will now give an example that may be very surprising.

7.3.11 Example The $\mathbb{Q}_1$ of rational numbers in $[0, 1]$ is a null set.

We enumerate $\mathbb{Q}_1 = \{r_1, r_2, \ldots\}$. Given $\varepsilon > 0$, note that the open interval $J_1 := (r_1 - \varepsilon/4, r_1 + \varepsilon/4)$ contains r_1 and has length $\varepsilon/2$; also the open interval $J_2 := (r_2 - \varepsilon/8, r_2 + \varepsilon/8)$ contains r_2 and has length $\varepsilon/4$. In general, the open interval

$$J_k := \left(r_k - \frac{\varepsilon}{2^{k+1}}, r_k + \frac{\varepsilon}{2^{k+1}}\right)$$

contains the point r_k and has length $\varepsilon/2^k$. Therefore, the union $\bigcup_{k=1}^{\infty} J_k$ of these open intervals contains every point of $\mathbb{Q}_1$; moreover, the sum of the lengths is $\sum_{k=1}^{\infty} (\varepsilon/2^k) = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\mathbb{Q}_1$ is a null set. □

The argument just given can be modified to show that: *Every countable set is a null set.* However, it can be shown that there exist uncountable null sets in $\mathbb{R}$; for example, the Cantor set that will be introduced in Definition 11.1.10.

We now state Lebesgue's Integrability Criterion. It asserts that a bounded function on an interval is Riemann integrable if and only if its points of discontinuity form a null set.

7.3.12 Lebesgue's Integrability Criterion *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if it is continuous almost everywhere on $[a, b]$.*

A proof of this result will be given in Appendix C. However, we will apply Lebesgue's Theorem here to some specific functions, and show that some of our previous results follow immediately from it. We shall also use this theorem to obtain the important Composition and Product Theorems.

7.3.13 Examples **(a)** The step function g in Example 7.1.4(b) is continuous at every point except the point $x = 1$. Therefore it follows from the Lebesgue Integrability Criterion that g is Riemann integrable.

In fact, since every step function has at most a finite set of points of discontinuity, then: *Every step function on $[a, b]$ is Riemann integrable.*

(b) Since it was seen in Theorem 5.6.4 that the set of points of discontinuity of a monotone function is countable, we conclude that: *Every monotone function on $[a, b]$ is Riemann integrable.*

(c) The function G in Example 7.1.4(d) is discontinuous precisely at the points $D := \{1, 1/2, \ldots, 1/n, \ldots\}$. Since this is a countable set, it is a null set and Lebesgue's Criterion implies that G is Riemann integrable.

(d) The Dirichlet function was shown in Example 7.2.2(b) not to be Riemann integrable.

Note that it is discontinuous at *every* point of $[0, 1]$. Since it can be shown that the interval $[0, 1]$ is not a null set, Lebesgue's Criterion yields the same conclusion.

(e) Let $h : [0, 1] \to \mathbb{R}$ be Thomae's function, defined in Examples 5.1.6(h) and 7.1.7.

In Example 5.1.6(h), we saw that h is continuous at every irrational number and is discontinuous at every rational number in $[0, 1]$. By Example 7.3.11, it is discontinuous on a null set, so Lebesgue's Criterion implies that Thomae's function is Riemann integrable on $[0, 1]$, as we saw in Example 7.1.7. □

We now obtain a result that will enable us to take other combinations of Riemann integrable functions.

7.3.14 Composition Theorem *Let $f \in \mathcal{R}[a, b]$ with $f([a, b]) \subseteq [c, d]$ and let $\varphi : [c, d] \to \mathbb{R}$ be continuous. Then the composition $\varphi \circ f$ belongs to $\mathcal{R}[a, b]$.*

Proof. If f is continuous at a point $u \in [a, b]$, then $\varphi \circ f$ is also continuous at u. Since the set D of points of discontinuity of f is a null set, it follows that the set $D_1 \subseteq D$ of points of discontinuity of $\varphi \circ f$ is also a null set. Therefore the composition $\varphi \circ f$ also belongs to $\mathcal{R}[a, b]$. Q.E.D.

It will be seen in Exercise 22 that the hypothesis that φ is continuous cannot be dropped. The next result is a corollary of the Composition Theorem.

7.3.15 Corollary *Suppose that $f \in \mathcal{R}[a, b]$. Then its absolute value $|f|$ is in $\mathcal{R}[a, b]$, and*

$$\left|\int_a^b f\right| \leq \int_a^b |f| \leq M(b - a),$$

where $|f(x)| \leq M$ for all $x \in [a, b]$.

Proof. We have seen in Theorem 7.1.6 that if f is integrable, then there exists M such that $|f(x)| \leq M$ for all $x \in [a, b]$. Let $\varphi(t) := |t|$ for $t \in [-M, M]$; then the Composition Theorem implies that $|f| = \varphi \circ f \in \mathcal{R}[a, b]$. The first inequality follows from the fact that $-|f| \leq f \leq |f|$ and 7.1.5(c), and the second from the fact that $|f(x)| \leq M$. Q.E.D.

7.3.16 The Product Theorem *If f and g belong to $\mathcal{R}[a, b]$, then the product fg belongs to $\mathcal{R}[a, b]$.*

Proof. If $\varphi(t) := t^2$ for $t \in [-M, M]$, it follows from the Composition Theorem that $f^2 = \varphi \circ f$ belongs to $\mathcal{R}[a, b]$. Similarly, $(f + g)^2$ and g^2 belong to $\mathcal{R}[a, b]$. But since we can write the product as

$$fg = \frac{1}{2}\left[(f + g)^2 - f^2 - g^2\right],$$

it follows that $fg \in \mathcal{R}[a, b]$. Q.E.D.

Integration by Parts

We will conclude this section with a rather general form of Integration by Parts for the Riemann integral, and Taylor's Theorem with the Remainder.

7.3.17 Integration by Parts *Let F, G be differentiable on $[a, b]$ and let $f := F'$ and $g := G'$ belong to $\mathcal{R}[a, b]$. Then*

$$\int_a^b fG = FG\Big|_a^b - \int_a^b Fg. \tag{7}$$

Proof. By Theorem 6.1.3(c), the derivative $(FG)'$ exists on $[a, b]$ and

$$(FG)' = F'G + FG' = fG + Fg.$$

Since F, G are continuous and f, g belong to $\mathcal{R}[a, b]$, the Product Theorem 7.3.16 implies that fG and Fg are integrable. Therefore the Fundamental Theorem 7.3.1 implies that

$$FG\Big|_a^b = \int_a^b (FG)' = \int_a^b fG + \int_a^b Fg,$$

from which (7) follows. Q.E.D.

A special, but useful, case of this theorem is when f and g are continuous on $[a, b]$ and F, G are their indefinite integrals $F(x) := \int_a^x f$ and $G(x) := \int_a^x g$.

We close this section with a version of Taylor's Theorem for the Riemann Integral.

7.3.18 Taylor's Theorem with the Remainder *Suppose that $f', \ldots, f^{(n)}, f^{(n+1)}$ exist on $[a, b]$ and that $f^{(n+1)} \in \mathcal{R}[a, b]$. Then we have*

$$f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + R_n, \tag{8}$$

where the remainder is given by

$$R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\cdot(b-t)^n dt. \tag{9}$$

Proof. Apply Integration by Parts to equation (9), with $F(t) := f^{(n)}(t)$ and $G(t) := (b-t)^n/n!$, so that $g(t) = -(b-t)^{n-1}/(n-1)!$, to get

$$\begin{aligned} R_n &= \frac{1}{n!}f^{(n)}(t)\cdot(b-t)^n\Big|_{t=a}^{t=b} + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\cdot(b-a)^{n-1}dt \\ &= -\frac{f^{(n)}(a)}{n!}\cdot(b-a)^n + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\cdot(b-t)^{n-1}dt. \end{aligned}$$

If we continue to integrate by parts in this way, we obtain (8). Q.E.D.

Exercises for Section 7.3

1. Extend the proof of the Fundamental Theorem 7.3.1 to the case of an arbitrary finite set E.
2. If $n \in \mathbb{N}$ and $H_n(x) := x^{n+1}/(n+1)$ for $x \in [a, b]$, show that the Fundamental Theorem 7.3.1 implies that $\int_a^b x^n dx = \left(b^{n+1} - a^{n+1}\right)/(n+1)$. What is the set E here?
3. If $g(x) := x$ for $|x| \geq 1$ and $g(x) := -x$ for $|x| < 1$ and if $G(x) := \frac{1}{2}\left|x^2 - 1\right|$, show that $\int_{-2}^3 g(x)dx = G(3) - G(-2) = 5/2$.
4. Let $B(x) := -\frac{1}{2}x^2$ for $x < 0$ and $B(x) := \frac{1}{2}x^2$ for $x \geq 0$. Show that $\int_a^b |x|dx = B(b) - B(a)$.
5. Let $f : [a, b] \to \mathbb{R}$ and let $C \in \mathbb{R}$.
   (a) If $\Phi : [a, b] \to \mathbb{R}$ is an antiderivative of f on $[a, b]$, show that $\Phi_C(x) := \Phi(x) + C$ is also an antiderivative of f on $[a, b]$.
   (b) If Φ_1 and Φ_2 are antiderivatives of f on $[a, b]$, show that $\Phi_1 - \Phi_2$ is a constant function on $[a, b]$.